1.4 Predicates and Quantifiers

Thursday, September 12, 2024 7:35 PM

We do not yet have all the tools to translate any phrase into logic.
We lack a method of extending and abstracting propositions as well as indicating quantity.

<u>Definition</u>: The <u>predicate</u> of a statement is a description of a property of the subject of the statement.

"$\underbrace{\text{Ohio}}_{\text{subject}}$ $\underbrace{\text{is a state.}}_{\text{predicate}}$"

"$\underbrace{x}_{\text{subject}} \underbrace{\geq 3}_{\text{predicate}}$"

<u>Definition</u>: A <u>propositional function</u> $P(x)$ is a statement with subject x which becomes a proposition once x is specified.

Let $P(x)$ denote "$x \geq 3$".
$P(3) \Rightarrow$ "$3 \geq 3$" which is (T).
$P(0) \Rightarrow$ "$0 \geq 3$" which is (F).

We might have more than one subject. In this case, our propositional function can look like

$$P(x_1, x_2, \dots, x_n).$$

Let $P(x, y, z)$ denote the statement $x^2 + y^2 = z$.
What is the truth value of
$P(1, 4, 17)$?

$P(1, 4, 17)$?

$P(0, 1, 2)$?

"Def" Quantifiers are used to refer to a range of elements. Think of words like "unique", "every", "none".

Definition: The <u>domain</u> of a prop function $P(x)$ is a specification of all x which may be considered when evaluating the truth of $P(x)$.

Let $P(x)$ be the statement "$2x^2 + 3x - 2 = 0$"

If the domain is only integers, $P(x)$ is only true when $x = -2$.

If the domain is all rational numbers $P(x)$ is true when $x = -2$ and $x = \frac{1}{2}$.

Let $P(x)$ be the statement "x is one of the 50 U.S. states."

If the domain is all states and provinces of Australia, $P(x)$ is never true.

Definition: The <u>universal quantification</u> of $P(x)$ is ...

"$P(x)$ for all x in the domain"

and denoted by $\forall x P(x)$.

Definition: The existential quantification of $P(x)$ is ...

"There exists an x in the domain such that $P(x)$."

and denoted by $\exists x P(x)$.

Definition: The uniqueness quantification of $P(x)$ is ..

<u>Definition</u>: The uniqueness quantification of P(x) is ...
"There exists a <u>unique</u> x in the domain such that P(x)."
and denoted by $\exists! x P(x)$.

Let the domain of x be all integers and determine the truth value of both

$$\forall x\,(x^2 > 0) \quad F,\ x=0 \Rightarrow x^2=0$$

$$\exists! x\,(x^2-4x+4=0) \quad T,\ x=2$$

Let O(x) be the statement
" x was born in the year 2004. "
where the domain is all of you in class. Express the following in words (but do not evaluate its truth)

$$\sim\exists x\, O(x)$$

There is not a student in class who was born in 2004.

$$\sim\forall x\, O(x)$$

Not every student in class was born in 2004.

Express the following in symbols..

There is a student in class who was not born in 2004.

$$\exists x\,(\sim O(x))$$

Every student in class was not born in 2004.

$$\forall x\,(\sim O(x))$$

Notice that some of these mean the same thing...

$$\boxed{\begin{array}{l} \sim\forall x\, O(x) \equiv \exists x\,(\sim O(x)) \\ \sim\exists x\, O(x) \equiv \forall x\,(\sim O(x)) \end{array}}$$

De Morgan's Law for Quantifiers